

 BRAC UNIVERSITY Inspiring Excellence	BRAC University Department of Computer Science and Engineering Semester: Fall 2024 Course Code: CSE460 Course Title: VLSI Design	Midterm Examination Full Marks: 10 x 2 + 15 = 35 Time: 1 hour 15 minutes Date: 22 November 2024
---	---	--

Set B

Student ID:	Name:	Section:
-------------	-------	----------

[Answer all **THREE** questions. Return the question paper with the answer script.]

1. (CO2) Consider a sequential circuit that detects if two consecutive bits are the same or not. The circuit generates an output $z=0$ if two consecutive bits are opposite and $z=1$ if the bits are the same. The output changes in the same clock cycle where the input is received.

w	0	0	1	0	0	1	1	0
z	0	1	0	0	1	0	1	0

Consider overlapping bit sequence

- Determine the type of FSM required for solving this problem. 1
- Draw the state diagram of the circuit. 3
- Derive a state-assigned table from (b) using the gray-encoding technique. 3
- Find the next-state and output logic expressions using k-map. 3

2. (CO2) Consider a sequential circuit with two inputs ($p1$ & $p2$) and a single output, z . The decisions regarding the status of various product orders are handled by two departments: the Sales Team and the Warehouse. Both departments independently approve (denoted by 1) or reject (denoted by 0) an order. The decisions of the Sales Team are given by a bit stream, $p1$, while the decisions of the Warehouse are represented by $p2$. If, for two consecutive orders, any one department rejects (i.e., **either $p1=0$ or $p2=0$ for two consecutive clock cycles**), the sequential circuit provides **$z=1$ at the next order**. The decisions of the Sales Team & the Warehouse for several orders and their corresponding synchronization instances are given below:

p1	1	0	0	1	0	1	1	0	1	0	1
p2	1	1	0	0	0	0	1	1	1	0	1
z	0	0	0	1	0	1	0	0	0	0	0

Note:

- You must use binary encoding to assign states.
- The overlapping of the input bits ($p1$ & $p2$) is not allowed.

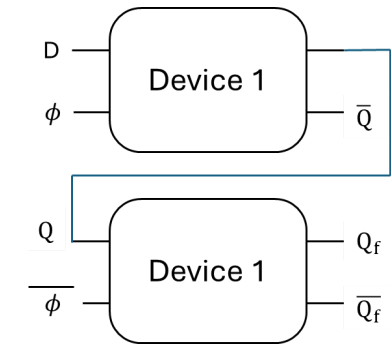
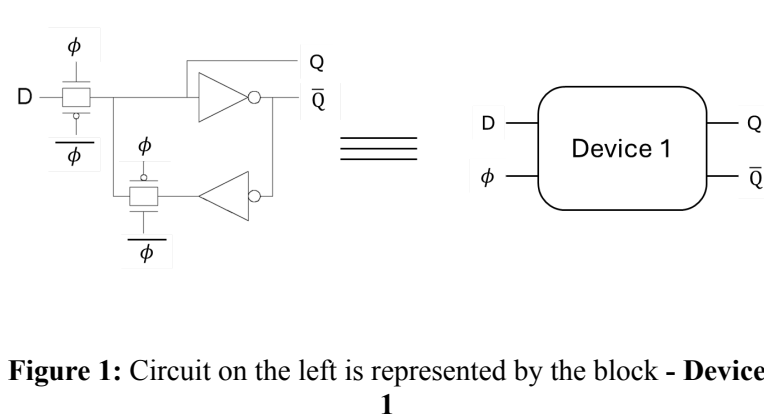
- Determine the type of FSM that would be needed for the given problem? 1
- Design the state diagram for the corresponding FSM 2
- Derive the state table and state-assigned table from your state diagram in (b). 3
- Evaluate the next state & output logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation). 4

3. (CO1) We have to implement an exam cheat detector. We have 4 sensors (A, B, C). Their functions are as follows:

- A:** High when a person has a cheatsheet in his/her pocket
- B:** High when a person has something written on his/her arms.
- C:** High when a person has AirPods hidden in his/her ears.

The output of the system will be high if the student should be expelled. The system is lenient enough to not expel if a student commits only offense **B** or only offense **C**. In any other case, the student will be expelled.

- (a) **Write** down the truth table for the system, describing all possible states of the sensors (A, B, C) 2
and their corresponding output states.
- (b) **Draw** the Karnaugh map (K-map) for the system based on the truth table in part (a). Derive 4
the logical expression that governs the system from the K-map.
- (c) **Draw** the CMOS logic circuit for the logical expression derived in part (b). Specify the number 3
of MOSFETs required to implement your design.



- (d) **Analyze** the circuit in Fig. 1 and write the truth table of the block **Device 1**. **Represent Q as a** 3
logic function of D when $\phi = 1$.
- (e) The table represents the time-dependent logical values (0 or 1) for each node in the circuit 3
depicted in Fig. 2, which is constructed using two Device 1 blocks. The Φ and D values are
provided in the table.
(X - don't care condition is a valid output too)

Node ↓ / Time →	5 ns	10 ns	15 ns	20 ns	25 ns
ϕ	0	1	0	1	0
D	0	1	1	1	0
Q					
Q_f	1				

Determine the values of **Q** and Q_f for the specified time intervals (5 ns, 10 ns, etc.). **Determine** if the circuit in Fig. 2 functions as a negative-edge/level-triggered or positive-edge/level-triggered flip-flop/latch?